

STANDARD

TRACEABILITY EDITION



BY METRASLATE

THE CALIBRATION RECORD FOR AI

PRODUCT DOCUMENT
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1. Summary

Organizations now make consequential decisions on the outputs of AI systems they did not build, cannot inspect, and do not control the updates of: a translation a call-taker acts on, a triage score a clinician sees, a risk rating a lender uses, a summary a caseworker relies on, a flag a supervisor receives. When one of those decisions is challenged, the organization is asked what the system produced, what the person saw, what the system had been validated for, whether it was performing to that validation at the time, and what changed since. Today none of that is a record. It is a reconstruction someone attempts from vendor dashboards, application logs, and memory.

The Traceability Edition of Standard is the record that answers those questions, built to the discipline calibration laboratories already use for instruments under accreditation. Every model version is an **instrument** with a **calibration record**: the reference cases it was evaluated against, per stratum; its validity domain; its uncertainty budget; the decision that placed it in service. Every consequential output is an **inference record** bound to the calibration record in force, the input's stratum and validity status, and an expanded uncertainty. A **validity gate** refuses to produce an output outside the validated domain and writes the refusal as a record. Reference cases are re-run continuously as control charts. A candidate model runs in shadow before promotion. When an instrument is found out of tolerance, every output produced under the invalid state is marked suspect and every decision and consequence that rested on it is enumerated by traversal of recorded lineage — the impact assessment a laboratory performs when a standard is found to have drifted. Because prior instruments' outputs are never overwritten, a change in a measured population is distinguishable from a change in its measurement, and a demographic disparity is located at the link of the chain where it enters before anything is called bias.

The edition ships three things: an **inference gateway** that sits between applications and models and is the enforcement point for binding and the gate; a **calibration program** — reference-set tooling, control charts, shadow deployment, promotion, out-of-tolerance handling, disparity testing — run on Standard's operations layer; and the **documents**: a hash-verified calibration certificate for every model version, an impact report for every out-of-tolerance event, and a reconstruction report for any consequential decision.

Standard is a domain-agnostic record and operations platform that ships as editions. It proved itself first in calibration laboratories, the record discipline that forgives the least, and its substrate — append-only records, hash-verified document reproduction, an exact-decimal value path, an uncertainty engine per JCGM 100, guard-banded decision rules per ILAC-G8, and fail-closed multi-tenant isolation — carries into this edition unchanged. The Traceability Edition is the reference implementation of the architecture described in MetraSlate's paper *Inference Traceability*, submitted as a public comment on NIST AI 200-2, and of the open specification of its record classes. The record names in this document are working names; the domain expert for each deployment replaces them.

2. The problem

2.1 Evaluation is an event; inference is continuous

Test, evaluation, verification, and validation of an AI system happens at a point in time — before deployment, at a periodic review, after an incident. The system then produces outputs continuously, on inputs the evaluation did not cover, under models the vendor updates on its own schedule, in conditions the evaluation did not stratify. Nothing binds each output to the state of the evaluation it was made under.



2.2 Nobody can say whether the people changed or the model did

A vendor promotes a new model version. The rate at which a system flags customers, patients, callers, or employees rises eighteen percent. The organization cannot tell whether the population changed, the inputs changed, the signal chain changed, the model changed, or the humans responding to the flags changed, because the prior model's outputs on the same evidence were never kept and no reference cases were re-run.

2.3 Confidence is not uncertainty, and a low score is not a refusal

Models produce a confidence value for any input, including inputs from populations, languages, and conditions the model was never validated for. A low confidence on such an input is presented as a measurement with wider error bars; it is an indication from an instrument used outside its specified range, and a laboratory would not report it at all.

2.4 Impact is an investigation, not a query

When a model is found to have been wrong for a population over an interval, the question that follows is which decisions rested on it. Answering it today means correlating application logs, vendor logs, case systems, and human memory. It takes weeks, it is incomplete, and it cannot be reproduced.

2.5 The obligations are arriving

The NIST AI Risk Management Framework's Measure and Manage functions; ISO/IEC 42001's monitoring and measurement requirements; the EU AI Act's record-keeping and post-market monitoring obligations for high-risk systems; the FDA's predetermined change control plans for AI-enabled device software; model risk management in banking; and U.S. state laws governing consequential automated decisions all require some form of what this edition records. None of them specifies the record. This edition does, and publishes the specification.

2.6 The costs

- **Litigation.** A decision challenged, and no defensible account of what the system produced, what the person saw, or whether the system was valid at the time.
- **Vendor changes.** A model update that silently shifts outcomes for a population, discovered months later by its consequences.
- **Bias findings without a locus.** A disparity reported at the outcome, with no way to say where in the chain it entered.
- **Wasted evaluation.** Pre-deployment evaluations that predicted nothing about post-deployment behavior, because no one measured the gap.
- **Unbounded recall.** An out-of-tolerance finding with no way to bound the affected decisions, so either everything is reviewed or nothing is.

3. The platform

Standard is one platform that ships as editions: the substrate — records, tenancy, documents, operations, numerics — is shared, and an edition declares its subjects, vocabulary, numbering, document templates, and reports in a manifest. The substrate carries no assumption about the industry it records; calibration was the first edition and the proving ground. The following are built.



The record spine. Records are never updated in place; a correction is a new record referencing the one it supersedes; the full history of any object is a read. The document engine is welded to the records that feed it: a rendered document carries a hash of its source data and a deterministic vocabulary projection, so a re-render years later either reproduces it exactly or fails verification. The value path carries exact decimals with explicit scale and never passes through floating point.

The metrology engine. Uncertainty propagation per JCGM 100 with effective degrees of freedom and coverage factors; decision rules with guard bands per ILAC-G8, with conformity, nonconformity, and the indeterminate zone as distinct results. Built for calibration certificates; used unchanged for inference uncertainty and the guard-banded decision rule.

The asset spine. Equipment with calibration due dates, service history, and the references each measurement traces to, with work orders and scheduling attached. In this edition the asset is the model: an instrument with a calibration interval, a due date, a service history, and references. The model registry is the asset spine.

The tenancy foundation. Row-level security enabled and forced on every tenant-scoped table; a connection without tenant context returns zero rows; dedicated databases for large tenants, a shared schema for the long tail, one placement layer between them; tenant context through asynchronous execution; an existence-and-count oracle in search found and closed.

The operations layer. Dispatch, role dashboards, the pipeline board, schedule and timecard, My Metrics, the persistent dock, personnel records, customer identity and book of business, in-context chat, the workflow engine.

Specified, next to build. The domain-event emission layer, which the gateway and adapters attach to, and systems-integrity monitoring, which turns control-chart state, coverage, and refusal rates into operational signals. Both are decided in the platform's decision record and compose primitives that exist.

The method. Standard is built by Metra Praxis: the domain expert owns the truth in her own words; every claim is tagged observed, claimed, or inferred; every fence is proven by a planted violation; a landing happens on a recorded word. The record's evidence grades are that epistemology applied to inferences.

4. The record model

4.1 Instrument

An instrument is a specific model version with a specific configuration and a specific input-conditioning pipeline — feature extraction, preprocessing, prompt template, retrieval configuration — identified by content hash. A change to any component is a new instrument. Instruments are assets on the asset spine: owner, custodian, calibration interval, due date, service history, validity status per stratum, and the references they are calibrated against.

4.2 Calibration record

The record establishing an instrument's relation to references: the reference set and version it was evaluated against; per-stratum results and control-chart state; the validity domain; the uncertainty budget; the coverage-check result; the promotion decision with its comparison report; and as-found and as-left states. Every inference record references the calibration record in force when it was produced; an inference without one is rejected at the writer.



4.3 Reference set

A curated, versioned corpus of reference cases, each with an expected output and the uncertainty of that expectation — the AI analogue of certified reference materials and check standards. Stratified across the conditions the instrument will encounter: for speech, language, dialect, speaker characteristics, microphone and codec, noise; for text, language, register, domain, length, source; for images, device, lighting, resolution. Each case carries its expected output, panel labels with inter-rater agreement recorded as label uncertainty, and the panel's composition. Sourced only where a lawful basis exists: consented recordings, licensed corpora, synthetic generation, the organization's own held data under its own terms. A sealed subset is never disclosed to model developers and is rotated; a fence asserts that no training pipeline can read it. A separate **conditioning reference set** — test signals through the actual input path — detects a change in the signal chain before it appears as a change in the model or the population.

4.4 Validity domain and status

The set of strata for which the calibration record holds reference evidence. Each stratum is *validated* (current evidence within tolerance and in control), *not validated* (no evidence), or *suspended* (a reference check failed or an investigation is open). Status is a record with history.

4.5 Uncertainty budget and decision rule

An inference carries an expanded uncertainty from a stated budget: conditioning repeatability from reference re-runs; model reproducibility across configurations and, for nondeterministic instruments, across repeated runs; input-quality contribution; validation coverage for the stratum; reference-label uncertainty. Combined per the GUM at a stated coverage factor. Any threshold that turns an inference into a categorical result — a flag, a routing, a conformity statement — is applied with a guard band: it fires only when the deviation exceeds the threshold by more than the expanded uncertainty; inside the guard band the result is *indeterminate* and recorded as such. A noisier measurement produces fewer flags, not more.

4.6 Inference record

One consequential output: input reference (a pointer, and a copy only where the register permits); measured input conditions and stratum; instrument; calibration record in force; validity status at the time; the output; the expanded uncertainty; the decision-rule result; timestamp; gateway identity and request identifier.

4.7 Presentation, decision, consequence

A **presentation record** states that an output was shown to a principal — which, when, in what surface — with a capture grade: *acknowledged*, *rendered*, or *available*. A **decision record** is a human or automated action with a *basis* (the presented outputs and other information it rested on, at a stated basis grade) and an *authority* (who or what had standing). A **consequence record** is a downstream event that references a decision as basis. Together with inference records they form a directed acyclic graph with basis edges, and lineage runs both ways: forward from an instrument to every consequence, backward from a consequence to the reference evidence its instrument rested on.

4.8 Refusal

The validity gate's decision not to infer, with reason codes: stratum not validated, stratum suspended, input quality below floor, conditioning reference out of control, instrument suspended, organization-wide suspension. Graded *absent with reason*. **NO VALID INFERENCE** is a row, never a blank.



4.9 Correction and discrepancy

A later human rendering, label, or ruling that disagrees with an output is a **correction** record referencing the output it corrects, marked post-hoc; the output is unchanged. Disagreement between two instruments on the same input — a vendor model and a reference rendering — is a **discrepancy** record with a lifecycle, never an override.

4.10 Commitments, retention, and lawful destruction

Every record is an envelope and a payload; the payload is content-addressed, encrypted under a per-tenant, per-class, per-period data key, and committed by a hash over the payload and a nonce stored only with it. A hash chain over envelopes seals the record. Retention is by record class; destruction deletes payload and nonce and destroys the key; the retained commitment proves the record existed without revealing it; derived records are found by reverse lineage and destroyed or re-derived. The instrument's longitudinal record survives lawful destruction of individual payloads.

4.11 The four decisions

Each data element is assigned *ingest*, *reference*, *derive*, or *never touch*, in that order of tests, in a versioned register the organization owns. Inputs are referenced by default and copied only where the register permits; features rather than raw inputs where extraction can run at the source; personal identifiers never resolved across records by the edition; a continuous minimization check reports any held payload whose justification has lapsed.

4.12 Evidence grades

Every statement carries *verified*, *received*, *derived*, *asserted*, or *absent* with reason. Absent is a value.

5. Ingest and integration

5.1 The inference gateway

The enforcement point. A service that sits between applications and models — vendor APIs, hosted services, local deployments — and, for every call: measures input conditions and assigns the stratum; evaluates the validity gate; forwards the call or writes a refusal; writes the inference record with the calibration record in force; returns the output with its record identifier, uncertainty, and decision-rule result. The gateway is stateless against Standard's record; it can run at the organization's edge or beside the model; it is the only path to the instrument in a complete deployment. Adapters for the major model APIs, for OpenAI-compatible endpoints, for on-premises inference servers, and for speech and translation vendors.

5.2 Coverage check

Where a vendor exposes usage logs, a scheduled reconciliation of vendor-reported calls against gateway records; unrecorded traffic is a finding reported by integrity monitoring and stated on every calibration record and impact report. Where no logs exist, the record states that coverage is unverified.

5.3 Presentation and decision capture

A client library for application surfaces that emits presentation records (rendered, acknowledged) with the inference identifier, and decision records with basis and authority. Where a surface cannot be instrumented,



presentation is recorded at *available* grade from delivery, and the record says so. Adapters for case-management, ticketing, and records systems supply consequence records by reference.

5.4 Reference renderings

Where a vendor exposes no model metadata, a second instrument under the organization's control produces a reference rendering at review time, graded derived, with its own calibration record; disagreement is a discrepancy record. The check standard beside the reading under test.

5.5 Model registries and pipelines

Adapters that register instruments from existing registries and CI pipelines, so a build that changes a model, a prompt template, or a preprocessing step produces a new instrument and a calibration requirement automatically, and a promotion in the pipeline is blocked until the promotion gate in Standard has passed.

5.6 What is not ingested

Training data. Model weights. Raw inputs beyond what the register permits. Personal identity resolved across records. The gateway forwards the call; it does not keep what the organization has no business keeping.

6. The calibration program

6.1 Reference-set management

Assemble, version, stratify, label, and seal. Panel labeling with inter-rater agreement and composition recorded. Lawful-basis attestation on every case. Coverage reporting: which strata the organization's live traffic presents, which the reference set covers, and the gap.

6.2 Continuous verification

Scheduled re-runs of the reference set and the conditioning reference set, per instrument, per stratum, charted as control charts. Out-of-control detection by ratified rules. Results written to the calibration record; state surfaced on the instrument board.

6.3 Shadow deployment and promotion

A candidate instrument receives live inputs beside the incumbent through the gateway, produces shadow inference records never presented, for a ratified period. The promotion gate compares per-stratum reference agreement, output rates on the same population, per-group rate change, and uncertainty distributions against ratified tolerances. Promotion is a recorded human decision with the comparison report as basis; the promoted instrument's first calibration record is its as-left state.

6.4 Out-of-tolerance handling

On a failed check or an upheld challenge: the stratum is suspended; every inference in the interval is marked suspect; the impact set is computed by forward traversal; downstream signals are withdrawn by record; affected principals are notified; the impact report is rendered and sealed; a root-cause record closes the event with a new calibration record.



6.5 Disparity across the chain

Comparison groups the organization defines and the classification register permits. Seven links measured — source data, conditioning, inference, threshold, flag, human response, outcome — with the first link at which groups diverge reported as a disparity with magnitude. Bias is a human finding after the investigation a disparity opens. Disparity series retained over time and intersected with calibration records and organizational events.

6.6 The kill switch

Per-stratum validity status is the gate; organization-wide suspension of an instrument is a recorded act with a reason; reinstatement is a recorded act with the evidence.

6.7 Calibration intervals

Every instrument has a calibration interval on the asset spine, set from its reference-set stability and the organization's policy, with a due date; an overdue instrument is a finding and, by policy, a suspension.

7. Documents

Rendered through the deterministic vocabulary projection and hash-verified against their records.

- **Model Calibration Certificate.** Per instrument version: identity and hash; reference set and version; per-stratum results and validity status; uncertainty budget; coverage-check result; promotion decision and comparison summary; as-found and as-left; interval and due date; the signatures of the calibration authority and the model owner.
- **Impact Report.** Per out-of-tolerance event: the instrument, the strata, the interval, the failed check, the suspect count, the impact set by decision type and consequence, notifications made, root cause, the new calibration record.
- **Reconstruction Report.** Per consequential decision: what the instrument produced, its uncertainty, what the person saw and at what capture grade, what was decided on what basis and authority, what followed, and the instrument's validity state at the time — every line graded.
- **Disparity Report.** Per period and comparison group: the seven links, the first divergence, the trend, and the organizational events within a stated lag.
- **Refusal Audit.** Per instrument and period: refusals by stratum and reason, against live traffic, so unvalidated demand is visible.

8. Operating the AI program

The instrument board. Every instrument in service with its validity status per stratum, control-chart state, coverage, refusal rate, calibration due date, and open out-of-tolerance events. What is drifting, what is suspended, what is overdue, what is refusing.

Pipeline. Instruments moving through registration, reference assembly, calibration, shadow, promotion, service, recalibration, retirement, with age-in-stage.

Schedule. Calibration intervals and due dates on the asset spine; reference re-runs on the schedule; shadow periods as scheduled work.



Roles. Model owner, calibration authority, reviewer, and the principals who act on outputs, each with a dashboard of what they own. My Metrics shows each reviewer their own findings; it never shows anyone a count of variances or refusals as a performance figure.

Workflow. Registration, promotion, out-of-tolerance response, challenge and review, and disparity investigation as agency-defined workflows with owners, deadlines, and a history on the record.

Review. As-known mode for any decision: what the person could have seen at the time, the instrument's state at the time, the later correction in a separate lane.

9. Scale and availability

Inference volume is the design driver. A large deployer produces millions of consequential inferences a day; each is one envelope with a small payload or a pointer. Inference records live in their own schema, partitioned by tenant, month, and record class, on the dedicated tier; append-only storage has no update churn; retention is a partition drop with extraction and key destruction. Structured search on tenant-leading indexes; no free-text search over payloads in this edition.

The gateway is on the request path and must not be the availability boundary of the application. It is horizontally scaled, adds single-digit milliseconds, and degrades by policy: if Standard's record is unreachable, the gateway either fails closed (no inference without a record — the default for consequential classes) or queues records durably and forwards (the policy for classes the organization designates), and the mode in force is itself recorded. Ingest is zero-loss behind a durable queue with seven days' retention. The platform targets 99.9 % monthly for interactive use, synchronous multi-AZ in-region, asynchronous cross-region disaster recovery.

10. Compliance program

- **Framework alignment.** NIST AI RMF Measure and Manage; ISO/IEC 42001 monitoring and measurement; EU AI Act record-keeping and post-market monitoring for high-risk systems; FDA predetermined change control plans; model risk management guidance; state automated-decision laws. The edition's documents are written to be the evidence each requires; the mapping is maintained per framework as a versioned record.
- **Platform assurance.** SOC 2 Type II as the baseline; HIPAA business associate path for PHI classes; CJIS through the Public Safety Edition's program; FedRAMP as a later program for federal deployers.
- **Consent and labor.** Any instrument that infers about employees requires the legal review, consent scopes, bargaining-unit terms, and purpose register the Public Safety Edition specifies, before it is registered.
- **Data-provider and model-provider terms.** Verified per adapter; they decide ingest versus reference per element and whether outputs may be retained.
- **The open specification.** Record classes and closure semantics published, PROV-serialized, with a conformance suite MetraSlate maintains and a conformance mark for passing implementations.



11. Technology

Kotlin on JDK 25 LTS with Spring Boot 4.1; sealed types for the record classes. Raw JDBC through a safe-SQL kit; no ORM. PostgreSQL 18, managed; RLS enabled and forced with class predicates on classified columns; leakproof-only operators on those columns; tenant-leading indexes. Content-addressed encrypted blob storage under a per-tenant, per-class, per-period key hierarchy. Deterministic vocabulary projection with hash verification for every document. Typed domain events, durable queue with idempotent replay, in-order writer. The gateway as a separate, horizontally scaled service with adapters as plugins and a client library for surfaces. Control charts and the uncertainty budget computed on the exact-decimal value path.

12. The method

Each deployment begins with Praxis Discovery, timeboxed, with the domain expert who acts on the instrument's outputs: the strata her traffic presents, the decision points and who decides, the purposes outputs may serve and the purposes they may not, the comparison groups her organization considers legitimate, the retention her rules require, and the off-limits list. Her words become the reference-set plan, the purpose register, and the classification register. The wedge is one instrument under calibration in thirty days with its acceptance audit written first. The build proceeds by fences with planted violations, refuters whose verdicts are reports, and landings on a recorded word.

13. Why this works

A calibration laboratory issues a certificate for an instrument, monitors it between calibrations, states the uncertainty of every result, decides conformity by a rule with a guard band, and recalls every result made on a standard that is found to have drifted. That discipline is codified, accredited, and older than the software industry. An AI model is an instrument. Standard was built to keep the laboratory's record, as a domain-agnostic platform that proved itself where records forgive the least; the Traceability Edition keeps the same record for the instrument that is now making the measurement, and publishes the format so that anyone can.

The proposition in one sentence, from the paper: an architecture for making the validity state of an AI system traceable to each consequential output, and for determining the downstream impact when that validity state later changes. This edition is that architecture as a product.